

AUTOMOTIVE INDUSTRY STANDARD

Medical Equipment for Two Wheeled Road Ambulances

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ON BEHALF OF
AUTOMOTIVE INDUSTRY STANDARDS COMMITTEE

UNDER
CENTRAL MOTOR VEHICLE RULES – TECHNICAL STANDING COMMITTEE

SET-UP BY
MINISTRY OF ROAD TRANSPORT and HIGHWAYS
GOVERNMENT OF INDIA

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INTRODUCTION

The Government of India felt the need for a permanent agency to expedite the publication of standards and development of test facilities in parallel when the work on the preparation of the standards is going on, as the development of improved safety critical parts can be undertaken only after the publication of the standard and commissioning of test facilities. To this end, the erstwhile Ministry of Surface Transport (MOST) has constituted a permanent Automotive Industry Standards Committee (AISC) vide order No. RT-11028/11/97-MVL dated September 15, 1997. The standards prepared by AISC will be approved by the permanent CMVR Technical Standing Committee (CMVR- TSC). After approval, the Automotive Research Association of India, (ARAI), Pune, being the Secretariat of the AIS Committee, will publish this standard. For better dissemination of this information ARAI may publish this document on their Web site.

During the deliberations, it was decided that type approval of vehicles shall be limited to automotive aspect of construction and functional requirements of life support two wheeled road ambulance and medical equipment's related details shall not be part of Vehicle Type Approval. 'Life Support – Two Wheeled road ambulance' standard is India unique and formulated based on the need for having two-wheeler ambulances which can have a significant usage in rural and semi urban areas. This Part 2 of the AIS-209 describes the specific requirements for Medical Equipment that shall be fitted to qualify it as "ambulance".

The AISC panel and the Automotive Industry Standards Committee (AISC) responsible for preparation of this standard are given in Annexure-1 and Annexure-2 respectively.

TABLE OF CONTENTS

Clause No	Description	Page No
1	Scope	1 / 7
2	Reference Standards	1 / 7
3	Definitions	1 / 7
4	General Requirements	1 / 7
5	Provision for Patients Safety and Medical Equipment's	3 / 7

LIST OF ANNEXUREURES

Annexure 1	Composition of AISC Panel	6 / 7
Annexure 2	Medical Equipment(s) for Ambulance Accessory Unit	7 / 7

Medical Equipment ⁽¹⁾ for Life Support Two Wheeled Road Ambulances

1.0 SCOPE

This standard specifies the guidelines regarding technical requirements for medical equipment which shall be installed in life support two-wheeled road ambulances for qualifying it as 'ambulance'. The concerned RTO to ensure that fitment of medical devices contained in this part in the ambulance accessory unit without compromising safety in consultation with committee of medical experts of the State.

2.0 REFERENCE STANDARDS

- 2.1 AIS-209 (Part 1) Construction and Functional Requirement For Life Support Two Wheeled Road Ambulance

3.0 DEFINITIONS

For the purpose of this standard, definitions given in AIS0-209 (Part 1) shall be referred. The following definition shall apply:

4.0 General Requirements

- 4.1 The Equipment should be designed for use in mobile situations and in field applications. If a medical Equipment is designated as "portable", which is meant for use inside an ambulance (except patient handling equipment according to Table 1) it should be in accordance with ISO/IEC 60601-1 and should

- a) Be possible to be carried by one person
- b) Have its own built in power supply (where relevant)
- c) Be capable of use outside the vehicle
- d) Be placed preferably along the street side wall of the patient compartment or along the ceiling ensuring the minimum possible distance to be connected to the patient without hindering the movement of personnel around the main stretcher

4.2 Temperature

(1) Medical equipment should be complaint with respective product standards as applicable (ISO/ BIS/ US FDA/ CE). AMC/ CMC and Calibration of the equipment shall be verified periodically by ambulance owner.

- 4.2.1 Unless otherwise marked on the Equipment, the Equipment should function as described in 4.2.2 and 4.2.3 when brought back to room temperature (20°C) after storage in temperatures ranging from - 30°C to 70°C.
- 4.2.2 Unless otherwise marked on the Equipment, the Equipment should function throughout the temperature range from 0°C to 40°C.

- 4.2.3 Unless otherwise marked on the Equipment, the Equipment should function for at least 20 min when placed in an environment at -5°C after storage at room temperature (20°C).

4.3 Humidity and ingress of liquids

Equipment's should comply with ISO/IEC 60601-1 and with particular Equipment standards of the series ISO/IEC 60601-2 where applicable.

4.4 Mechanical strength

4.4.1 General

Where there are not more stringent requirements for mechanical strength in particular Equipment's' standards, then the following mechanical strength requirements should apply to medical Equipment's for use in road ambulances.

4.4.2 Vibration and bump

After vibration tests and bump test in accordance with 4.11 the maintain system and Equipment should function within the tolerances specified by the manufacturer.

4.4.3 Free fall

If the medical Equipment is fixed, as defined in ISO/IEC 60601-1 it is exempted from the free fall test.

Equipment's which are taken out of holders and/or carried by hand should be submitted to the free fall test according 4.12 and should then function within the tolerances specified by the manufacturer.

Note: A medical Equipment may consist of fixed and loose components, the free fall test applies to the loose components only.

4.5 Fixation of Equipment's

- 4.5.1 The fixation system(s), maintain system(s) or storage system(s) should hold the Equipment to withstand accelerations or decelerations of 10 g longitudinal (forward, backward), 10 g transverse (left, right) and 10 g vertical.

Terminal units and electrical socket outlets should not be used as part of the fixation system.

If rails systems are used, they should comply with ISO 19054.

Note: Rail systems consist of e.g. rail supports, rails, rail clamps, equipment mount holders, equipment mounts, equipment pin holders and equipment pins.

4.6 Electrical safety

All Equipment's should be selected and mounted so that no harmful influence to the electrical supplies results.

4.7 User interface

Buttons, switches, indicators and controls should be easily accessible and visible. SI units (except for blood pressure and airway pressure) and standardized graphical symbols where applicable should be used.

5.0 PROVISION MEDICAL EQUIPMENTS

5.1 Life support Two wheeled Road Ambulance shall have medical equipment's as given in Table 1 below.

Table 1 Medical Equipment(s) for Ambulance Accessory Unit ⁽¹⁾		
S.No.	Equipment	Requirement(s)
1.	Main Stretcher/ Under Carriage ⁽²⁾	01
2.	Cervical Upper Spinal Immobilization Equipment ⁽²⁾	01
3.	Multipurpose malleable splint with compression bandage set	01
4.	Portable oxygen ⁽²⁾	01
5.	Extended Support oxygen (Type B Cylinder) ⁽³⁾	01
6.	Valve for Cylinders 1	01
7.	Resuscitator or with oxygen inlet and masks and airways for all ages and oxygen reservoir	01
8.	Mouth to mask ventilator with oxygen inlet	01
9.	Electrical or pedal Portable Aspirator & its manual	01
10.	Stationary Oxygen (Min 10 LPM oxygen concentrator) ⁽⁴⁾	01
11.	Pain Relief	X
12.	Infusion Solution 1 Litre	02
13.	Material for Treatment of Wounds	01
14.	Tourniquet and bleed controlling Equipment's kit (clotting agents & bandages etc).	01
15.	Material for Treatment of Burns and corrosives	01
16.	Vomiting Bag	02
17.	Non-glass urine bottle	01
18.	Sharps container	01
19.	Sterile Surgical Gloves, Pairs	05
20.	Non-Sterile Gloves for Single Use	100
21.	Emergency Delivery Kit ⁽³⁾	01
22.	Waste Bag ⁽⁵⁾	01
23.	Clinical Waster Bag (color coded yellow, blue and	15 (05 each)

	black)	
24.	Non-woven stretcher sheet	01
25.	PPE Kit (as per standard government norms) for daily use and/or infection control as per site of deployment.	02
26.	Cleaning and disinfection material ⁽²⁾	01
27.	Rescue tools	X
28.	Seat belt cutter	01
29.	Warning Triangle Lights	02
30.	Fire Extinguisher, ABC Type (min 2 Kg capacity) ⁽²⁾	01
31.	Mobile phone dedicated with Ambulance in sync with fitted GPS Equipment	01
32.	Internal communication between patient compartment and outside ⁽²⁾	01
⁽¹⁾ All the items shall be kept in inventory for continuous replacement for daily/ weekly usage with replacement Equipment's available ensuring minimum response time during emergencies.		
⁽²⁾ Specifications and positioning is mentioned in AIS-209		
⁽³⁾ b Optional as per space availability;		
⁽⁴⁾ Essential supply for which fitment is design to use it during emergency while using ambulance accessory unit as standalone unit; 'X' requirements refers to site of deployment or user's specific requirement.		
⁽⁵⁾ Waste bag to be a Biohazard bag. Waste generated to be disposed as per applicable Biomedical waste handling norms		

ANNEXURE 1

(See Introduction)

**COMPOSITION OF AISC PANEL ON
CONSTRUCTION AND FUNCTIONAL REQUIREMENT FOR LIFE
SAVING-TWO WHEELED ROAD AMBULANCE**

Name	Organization
Mr. Amit Gupta (Convener)	All India Institute of Medical Sciences
Mr. A. A. Badusha	ARAI
Mr. Vikram Tandon	ARAI
Mr. Kamalesh B. Patil	ARAI
Mr. Pratik Nayak	ARAI
Mr. Sujit Chinchwade	ARAI
Mr. S. N. Dhole	CIRT
Mr. Deepak Joshi	ICAT
Mr. Gurkaran Singh	ICAT
Mr. Deepak Joshi	ICAT
Mr. Nagarajan S	GARC
Dr. Kailash Manda	---
Dr. Raman Chawla	---
Mr. K.Kapila	IRF
Mr. Balraj Bhanot	IAC & IRF
Mr. Parmeshwar Mane	SIAM (Bajaj Auto Ltd.)
Mr. Piyush Chowdhry	SIAM (Hero Moto. Corp. Ltd.)
Mr. Ishan Gupta	SIAM (Hero Moto. Corp. Ltd.)
Mr. Shiva Arun Teja	SIAM (Hero Moto. Corp. Ltd.)
Mr. Apoorva Tripathi	SIAM (Hero Moto. Corp. Ltd.)
Ms. Garima	SIAM (Hero Moto. Corp. Ltd.)
Mr. S. Gururajan	SIAM (TVS Motor Company Ltd.)
Mr. Sanchit Khare	OLA Electric
* At the time of approval of this Automotive Industry Standard (AIS)	

ANNEXURE 2 (See Introduction) COMMITTEE COMPOSITION * Automotive Industry Standards Committee	
Chairperson	
Dr. Reji Mathai	Director, The Automotive Research Association of India
Members	Representing
Representative from	Ministry of Road Transport and Highways
Representative from	Ministry of Heavy Industries
Representative from	Office of the Development Commissioner, MSME, Ministry of Micro, Small and Medium Enterprises
Shri Shrikant R. Marathe	Former Chairman, AISC
Head-TED	Bureau of Indian Standards
Director	Central Institute of Road Transport
Director	Global Automotive Research Centre
Director	International Centre for Automotive Technology
Director	Indian Institute of Petroleum
Director	National Automotive Test Tracks
Director	Vehicles Research and Development Establishment
Director	Indian Rubber Manufacturers Research Association
Representatives from	Society of Indian Automobile Manufacturers
Representatives from	Tractor and Mechanization Association
Representatives from	Automotive Components Manufacturers Association of India
Representative from	Indian Construction Equipment Manufacturers' Association
Member Secretary	
Shri Vikram Tandon	The Automotive Research Association of India

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